Changying Ding

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Academic Employment

Lecturer, University of California, Los Angeles, 2026–2027.

Hedrick Math Fellow, University of California, Los Angeles, 2023–2026.

Mentors: Sorin Popa and Dimitri Shlyakhtenko.

Education

Ph.D. in Mathematics, Vanderbilt University, USA, May 2023.

Advisor: Jesse Peterson.

B.S. in Mathematics, Central South University, China, May 2018.

Research Interests

Operator algebras (von Neumann and C*-algebras), ergodic theory (measure and orbit equivalence), and analytic and geometric group theory.

Grants

NSF Grant DMS-2554483, 2026–2029.

AMS-Simons Travel Grant, 2024–2026.

Publications/Preprints

9. C. Ding, Z. Yang, *Non-isomorphism of q-Gaussian von Neumann algebras in infinite variables*, arXiv:2609.07052.
8. C. Ding, H. Tan, *Structural and non-isomorphism results for q-Araki-Woods factors*, arXiv:2509.21636, **Journal of Functional Analysis**, 291 (2026), no. 5, 111541.
7. C. Ding, D. Drimbe, *Relative solidity for biexact groups in measure equivalence*, arXiv:2503.24167.
6. C. Ding, *A unique Cartan subalgebra result for Bernoulli actions of weakly amenable groups*, arXiv:2404.08182, **Journal of Functional Analysis**, 288 (2025), no. 10, 110852.
5. C. Ding, J. Peterson, *Biexact von Neumann algebras*, arXiv:2309.10161.
4. C. Ding, *First ℓ^2-Betti numbers and proper proximality*, arXiv:2211.05951, **Advances in Mathematics**, 438 (2024), 109467.

3. C. Ding, S. Kunnawalkam Elayavalli, *On the structure of relatively biexact group von Neumann algebras*, arXiv:2211.05298,
Communications in Mathematical Physics, 405 (2024), article 104.

2. C. Ding, S. Kunnawalkam Elayavalli, J. Peterson, *Properly proximal von Neumann algebras*, arXiv:2204.00517,
Duke Mathematical Journal, 172 (2023), no. 15, 2821–2894.

1. C. Ding, S. Kunnawalkam Elayavalli, *Proper proximality for various families of groups*, arXiv:2107.02917,
Groups, Geometry, and Dynamics, 18 (2024), no. 3, 921–938.

Talks

Great Plains Operator Theory Symposium, University of Iowa, May 2026.

Functional Analysis Seminar, University of California, San Diego, April 2026.

Mini-school around Sofic Groups and the Connes Embedding Problem, UT Austin, March 2026.

Recent Development in Operator Algebras, University of Maryland, February 2026.

Colloquium, University of Iowa, November 2025.

East Coast Operator Algebras Symposium, University of Nebraska-Lincoln, October 2025.

Subfactor Seminar, Vanderbilt University, April 2025.

Functional Analysis Seminar, University of California, San Diego, April 2025.

Operator Theory Seminar, University of Iowa, November 2024.

Group Operator Algebras: Classification, Structure and Rigidity, BIRS, September 2024.

Noncommutative Geometry in NYC, May 2024.

Subfactor Seminar, Vanderbilt University, April 2024.

Functional Analysis Seminar, Oxford University, March 2024.

Functional Analysis Seminar, University of California, San Diego, December 2023.

Analysis Seminar, University of Houston, November 2023.

Probabilistic Operator Algebra Seminar, University of California, Berkeley, October 2023.

Workshop on Operator Algebras and Applications: Groups and Group Actions, Fields Institute, October 2023.

Seminar in Operator Theory and Operator Algebras, University of Virginia, September 2023.

Noncommutative Geometry and Operator Algebras Spring Institute, Vanderbilt University, May 2023.

Workshop on Operator Algebras and Quantum Information, University of Delaware, May 2023.

Workshop III in Von Neumann Algebras and Geometric Group Theory, University of Iowa, April 2023.

Operator Algebras Seminar, Purdue University, March 2023.

Workshop on Von Neumann Algebras and Geometric Group Theory, University of California, San Diego, February 2023.

AMS Special Session on Advances in Operator Algebras, Joint Mathematics Meetings, January 2023.

Functional Analysis Seminar, University of California, Los Angeles, November 2022.

East Coast Operator Algebras Symposium, Michigan State University, October 2022.

Operator Theory Seminar, University of Iowa, October 2022.

Great Plains Operator Theory Symposium, Washington University in St. Louis, May 2022.

Functional Analysis Seminar, University of California, San Diego, May 2022.

Service

Co-organizer, AMS Special Session on Advances in Operator Algebras, Joint Mathematics Meetings, 2026.

Co-organizer, AMS Special Session on Advances in Operator Algebras, Joint Mathematics Meetings, 2025.

Co-organizer, Functional Analysis Seminar, University of California, Los Angeles, 2023–present.

Journals refereed: Adv. Math., Analysis & PDE, Duke Math. J., Forum Math. Sigma, Infin. Dimens. Anal. Quantum Probab. Relat. Top., Pacific J. Math., Proc. Amer. Math. Soc., Proc. Roy. Soc. Edinburgh Sect. A, Trans. Amer. Math. Soc.

Teaching

Instructor — Fall 2025
MATH 31A Differential and Integral Calculus — University of California, Los Angeles

Instructor — Winter 2024, Fall 2024
MATH 31B Integration and Infinite Series — University of California, Los Angeles

Instructor — Fall 2023, Winter 2024, Spring 2024, Winter 2025, Winter 2026
MATH 131A Analysis — University of California, Los Angeles

Instructor — Fall 2024, Spring 2025, Winter 2026, Spring 2026
MATH 131B Analysis — University of California, Los Angeles